

Kiera McCormick — CV

Johns Hopkins University

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RESEARCH INTERESTS

My research focuses on large language models for scientific and real-world understanding, including retrieval-augmented generation, prompt engineering, and question-answering systems. I am particularly interested in human–AI comparison and cooperation on natural language processing tasks, evaluation of human–computer interaction in high-stakes or scientific settings, and benchmarking LLM behavior. More broadly, my interests include open-source science, interdisciplinary applications of machine learning, machine translation, and AI safety.

EDUCATION

Johns Hopkins University

Baltimore, MD

Master of Science in Computer Science

Expected Graduation: May 2027

- **Concentration:** Human Language Technology
- **Relevant Coursework:** Natural Language Processing, Human Language Technology

Loyola University Maryland

Baltimore, MD

Bachelor of Science in Engineering

Graduated May 2025

- **Concentrations:** Electrical and Computer Engineering
- **Minor:** Mathematics
- **GPA:** 3.745
- **Dean's List Honors:** Fall 2021, Spring 2022, Fall 2022, Spring 2023, Fall 2023, Spring 2024, Fall 2024, Spring 2025
- **Relevant Coursework:** Physics I & II, Linear Circuit Analysis, Statics, Calculus I, II, III, Electronics, Digital Logic, Quantum Computing, Signals and Systems, Ordinary Differential Equations, Programming Tools, Object-Oriented Engineering Design, Probability and Statistics, FPGA Design, Communications, Linear Algebra, Electromagnetics, Microprocessor-Based Systems, Advanced Linear Algebra, Quantum Computing II, Power Systems, Engineering Systems Analysis

Danish Institute for Study Abroad

Copenhagen, DK

Core Course: Holocaust and Genocide

Spring 2024

RESEARCH AND WORK EXPERIENCE

Center for Astrophysics | Harvard & Smithsonian

Cambridge, MA

AstroAI Summer Intern

January 2025 – August 2025

- Engineered prompts to generate concise and discriminative summaries about astrophysical X-ray sources from Chandra Archives data and created visualizations to understand the prompting effects.
- Extracted embedded physical parameters to evaluate LLMs' understanding of scientific knowledge.
- Applied sparse autoencoders to reveal novel patterns and relationships among astrophysical objects.

Space Telescope Science Institute and Johns Hopkins University

Baltimore, MD

Summer Astronomy Space Program and the Annual Jelinek Memorial Summer Workshop

May 2024 – August 2024

- Examined the application of artificial intelligence and machine learning, particularly Large Language Models (LLMs), in astronomy research. Assessed the potential of LLMs to enhance research efficiency and reliability.
- Designed and implemented LLM-powered chatbots for deployment on Slack, specializing in prompt engineering, conducting comparisons between open and closed-source models, integrating information retrieval pipelines, and creating evaluation benchmark datasets.

Omega Technical Services at Los Alamos National Laboratory

Los Alamos, NM

Engineering Intern

May 2023 – August 2023

- Collaborated with the industrial engineering team at the Los Alamos National Laboratory, focusing on task scheduling optimization through databases and automation. Transformed a previously manual task of job scheduling to an automated system.
- Utilized database coding and conducted data analysis using tools such as MicroStrategy, Excel, SQL, Python, VBA, and Confluence, contributing to the creation of metrics and dashboards for effective data visualization.
- Worked towards automating an extract, transform, load (ETL) pipeline through MariaDB, MS Task Scheduler, and Alteryx.

Loyola University Maryland

Baltimore, MD

Physics Teaching Assistant and Tutor

September 2022 – May 2025

- Supported undergraduate instruction in Physics I and II through problem-solving sessions and individualized tutoring, emphasizing quantitative reasoning and conceptual understanding.

University of Delaware

Newark, DE

Junior Physics Intern

March 2020 – August 2020

- Contributed to an astrophysical research project analyzing the spectral characteristics and luminosity of massive stars. Developed Python scripts using Jupyter Notebook to process and visualize complex stellar data.
- Created data visualizations of spectral lines and magnitude measurements, enhancing the interpretation of stellar properties.

PUBLICATIONS

- [*Encoding and Understanding Astrophysical Information in Large Language Model-Generated Summaries*](#)
 - **McCormick, K.**, Martínez-Galarza, J. R., 2025, *NeurIPS: ML4PS*, accepted.
- [*From Queries to Criteria: Understanding How Astronomers Evaluate LLMs*](#)
 - **McCormick, K.**, Hyk, A., Zhong, M., Ciucă, I., et al., 2025, *COLM*, accepted.
- [*Designing an Evaluation Framework for Large Language Models in Astronomy Research*](#)
 - Wu, J. F., Hyk, A., **McCormick, K.**, Ye, C., et al., 2024, *ICML: AI4Science workshop*, submitted, arXiv:2405.20389.
- [*pathfinder: A Semantic Framework for Literature Review and Knowledge Discovery in Astronomy*](#)
 - Iyer, K. G., Yunus, M., O'Neill, C., Ye, C., Hyk, A., **McCormick, K.**, et al., 2024, *ApJS* 275 38, arXiv:2408.01556.

CONFERENCES

Machine Learning conference/workshop paper reviewer for *NeurIPS* for the EvalEval Workshop in 2024.

Volunteer at the *AstroAI Workshop*

Gender Minorities and Women in Physics Summit

Johns Hopkins University

Poster Presentation

September 14, 2024

- Discussed work on the application of Large Language Models in Physics and Astronomy.

American Astronomical Society 245

National Harbor, MD

iPosters

January 11-16, 2025

- Presented “Evaluating Large Language Models in Astronomy Research” at the Machine Learning Methods session.
- Co-authored “pathfinder: a Semantic Framework for Literature Review and Knowledge Discovery in Astronomy” at the Machine Learning Methods session.

Talks

- Co-authored “Evaluating LLM Tools: Insights from Astronomer Interactions with a RAG-Based Slack Chatbot” at the Technology Developments in Outreach, Education, and Research session.

AstroAI Workshop

Cambridge, MA

Poster Presentation

July 7-11, 2025

- Presented “Evaluating Large Language Models in Astronomy Research”.

Conference on Language Modeling

Québec, Canada

Poster Presentation

October 7-10, 2025

- Presented “From Queries to Criteria: How Astronomers Evaluate LLMs”.

Conference on Neural Information Processing Systems

San Diego, California

Poster Presentation

December 2-7, 2025

- Presented “Encoding and Understanding Astrophysical Information in Large Language Model-Generated Summaries” at the Machine Learning for the Physical Sciences Workshop.

Upcoming

STScI Language AI in the Space Sciences

Baltimore, MD

Poster Presentation

March 9-12, 2026

- Presenting “Encoding and Understanding Astrophysical Information in Large Language Model-Generated Summaries”.

TECHNICAL & RESEARCH SKILLS

- **Programming:** Python, SQL/SQLite, Java, C++, Git/GitLab, HTML
- **ML/AI:** familiarity with PyTorch and TensorFlow, LLMs, retrieval-augmented generation and information retrieval, representation learning, model evaluation and benchmarking, prompt engineering
- **Scientific/Engineering tools:** MATLAB (certified), Simulink, LT Spice, Active HDL, MicroStrategy
- **Hardware/Systems:** Raspberry Pi, Arduino, FPGA, embedded systems

Completion of the 2024 Summer School on Human Language Technology at Johns Hopkins University

SELECTED PROJECTS

Hotspot Homing Robot

September 2024 – May 2025

- Developed an autonomous mobile robot designed to detect and localize rogue access points in office environments, sponsored and mentored by members of the Applied Signal Technology sector at Raytheon.
- Integrated wireless signal processing and detection techniques with navigation systems, such as Simultaneous Localization and Mapping, to create a comprehensive security solution.
- Collaborated with an interdisciplinary team to design a user-friendly interface for real-time threat visualization and reporting.

EXTRACURRICULAR ACTIVITIES

Johns Hopkins Women's Club Lacrosse Team

Member

Baltimore, MD

September 2025 – Present

Society of Women Engineers

President

Baltimore, MD

February 2023 – May 2025

Engineering Industrial Advisory Board

Class of 2025 Representative

Baltimore, MD

December 2023 – May 2025

Loyola University Maryland Honors Program

Member and Peer Mentor

Baltimore, MD

September 2021 – May 2025

Haig Scholar

Member

Baltimore, MD

February 2024 – May 2025

Loyola Women's Club Lacrosse Team

Academic All-American

Baltimore, MD

September 2022 – May 2025

Institute of Electrical and Electronics Engineers

Member

Baltimore, MD

October 2021 – May 2025